

10.1 Convergent and Divergent Infinite Series

Calculus

Practice

1. Given the infinite series $\sum_{n=1}^{\infty} (-1)^n$, find the sequence of partial sums S_1, S_2, S_3, S_4 , and S_5 .

$$\begin{aligned} S_1 &= -1 & S_3 &= S_2 - 1 = -1 & S_5 &= S_4 - 1 = -1 \\ S_2 &= S_1 + 1 = 0 & S_4 &= S_3 + 1 = 0 \end{aligned}$$

2. Find the sequence of partial sums S_1, S_2, S_3, S_4 , and S_5 for the infinite series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} + \frac{1}{10} + \dots$.

$$\begin{aligned} S_1 &= 1 & S_3 &= \frac{3}{2} + \frac{1}{4} = \frac{7}{4} & S_5 &= \frac{23}{12} + \frac{1}{8} = \frac{49}{24} \\ S_2 &= \frac{3}{2} & S_4 &= \frac{7}{4} + \frac{1}{6} = \frac{23}{12} \end{aligned}$$

3. If the infinite series $\sum_{n=1}^{\infty} a_n$ has n th partial sum $S_n = (-1)^{n+1}$ for $n \geq 1$, what is the sum of the series?

$$\lim_{n \rightarrow \infty} S_n \quad \boxed{\text{diverges}} \quad 1, -1, 1, -1, \dots$$

4. The infinite series $\sum_{n=1}^{\infty} a_n$ has n th partial sum $S_n = \frac{n}{4n+1}$ for $n \geq 1$. What is the sum of the series?

$$\lim_{n \rightarrow \infty} \frac{n}{4n+1} = \frac{1}{4}$$

5. Use a calculator to find the partial sum S_n of the series $\sum_{n=1}^{\infty} \frac{6}{n(n+3)}$ for $n = 100, 500, 1000$.

$$S_{100} \approx 3.6078 \quad S_{500} \approx 3.6547 \quad S_{1000} \approx 3.6606$$

6. Show that the sequence with the given n th term $a_n = 1 + 2n$ is monotonic.

$$a_1 = 3 \quad a_2 = 5 \quad a_3 = 7 \quad a_4 = 9 \quad \dots$$

$$a_n \text{ is monotonic because } a_1 \leq a_2 \leq a_3 \leq \dots \leq a_n$$

7. What is the n th partial sum of the infinite series $\sum_{n=1}^{\infty} \frac{1}{2^{n+1}}$?

$$\begin{aligned} S_n &= \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots + \frac{1}{2^{n+1}} \\ \frac{1}{2} S_n &= \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots + \frac{1}{2^{n+2}} \\ S_n - \frac{1}{2} S_n &= \frac{1}{2^2} - \frac{1}{2^{n+2}} \end{aligned}$$

$$\begin{aligned} \frac{1}{2} S_n &= \frac{1}{2^2} - \frac{1}{2^{n+2}} \\ S_n &= 2 \left[\frac{1}{2^2} - \frac{1}{2^{n+2}} \right] \end{aligned}$$

$$\boxed{S_n = \frac{1}{2} - \frac{1}{2^{n+1}}}$$

10.1 Convergent and Divergent Infinite Series

Test Prep

8. Which of the following could be the n th partial sum for the infinite series $\sum_{n=1}^{\infty} \frac{1}{4^n}$?

$$\begin{aligned}
 S_n &= \frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \dots + \frac{1}{4^n} \\
 \frac{1}{4} S_n &= \frac{1}{4^2} + \frac{1}{4^3} + \frac{1}{4^4} + \dots + \frac{1}{4^{n+1}} \\
 S_n - \frac{1}{4} S_n &= \frac{1}{4} - \frac{1}{4^{n+1}}
 \end{aligned}$$

$$\begin{aligned}
 \frac{3}{4} S_n &= \frac{1}{4} - \frac{1}{4^{n+1}} \\
 3 S_n &= 1 - \frac{1}{4^n} \\
 S_n &= \frac{1}{3} \left(1 - \frac{1}{4^n} \right)
 \end{aligned}$$

- (A) $S_n = \frac{1}{3} \left(1 + \frac{1}{4^n} \right)$ (B) $S_n = \frac{1}{3} \left(1 - \frac{1}{4^{n+1}} \right)$ (C) $S_n = \frac{1}{3} \left(1 - \frac{1}{4^n} \right)$ (D) $S_n = \frac{1}{4} \left(1 - \frac{1}{3^n} \right)$

9. If the infinite series $\sum_{n=1}^{\infty} a_n$ is convergent and has a sum of $\frac{7}{8}$, which of the following could be the n th partial sum?

$$\lim_{n \rightarrow \infty} S_n = \frac{7}{8}$$

(A) $S_n = \frac{7n+1}{8n^2+1} \rightarrow 0$

(B) $S_n = \frac{7n^2+1}{8n+1} \rightarrow \infty$

(C) $S_n = 2 \left(\frac{7}{8} - \frac{1}{n+2} - \frac{1}{n+3} \right) \rightarrow \frac{14}{8}$

(D) $S_n = \left(\frac{7}{8} - \frac{1}{n+2} - \frac{1}{n+3} \right) \rightarrow \frac{7}{8}$

10. Which of the following sequences with the given n th term is bounded and monotonic?

Bounded,
not monotonic



(A) $a_n = 2 + (-1)^n$

not bounded,
monotonic



(B) $a_n = \frac{n^2}{n+1}$

bounded,
monotonic



(C) $a_n = \frac{3n}{n+2}$

bounded,
not monotonic

(D) $a_n = \frac{\cos n}{n}$